

Context-Aware and Efficient Data Valuation for Medical Machine Learning

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Background

Medical imaging datasets are rapidly growing, but more data does not always lead to better models [1]. Redundant, mislabeled, or low-quality images can slow training, waste resources, and even degrade performance.

- These challenges are amplified in medical imaging:
- Annotation is expensive (requires clinical experts)
 - Data often contains redundancy and noise [2,3]

Thus, simply increasing dataset size is inefficient and costly. Data valuation assigns an importance score to each data point based on its impact on model performance

- High-value: essential for learning
- Low-value: redundant or noisy
- Negative-value: harmful to performance

Focusing on high-value data enables:

- Faster and more efficient training
- Reduced computational and time costs
- Maintained model performance with minimal loss [1]

Ultimately, data valuation shifts the focus from more data to better data, enabling more efficient, sustainable, and trustworthy machine learning in healthcare.

Goal

Build an **open source GitHub** project for **ML researchers** investigating **the reduction of data** through **data valuation** methods to train medical imaging ML models while **retaining model performance** through data valuation techniques.

Dataset and ML Models

Task	Data Set	ML Model	Evaluation Metric
X-ray Classification	NIH Chest X-ray [4]	DenseNet 121	AUROC
MRI Segmentation	BraTS-2023-GLI [5]	U-net	Dice

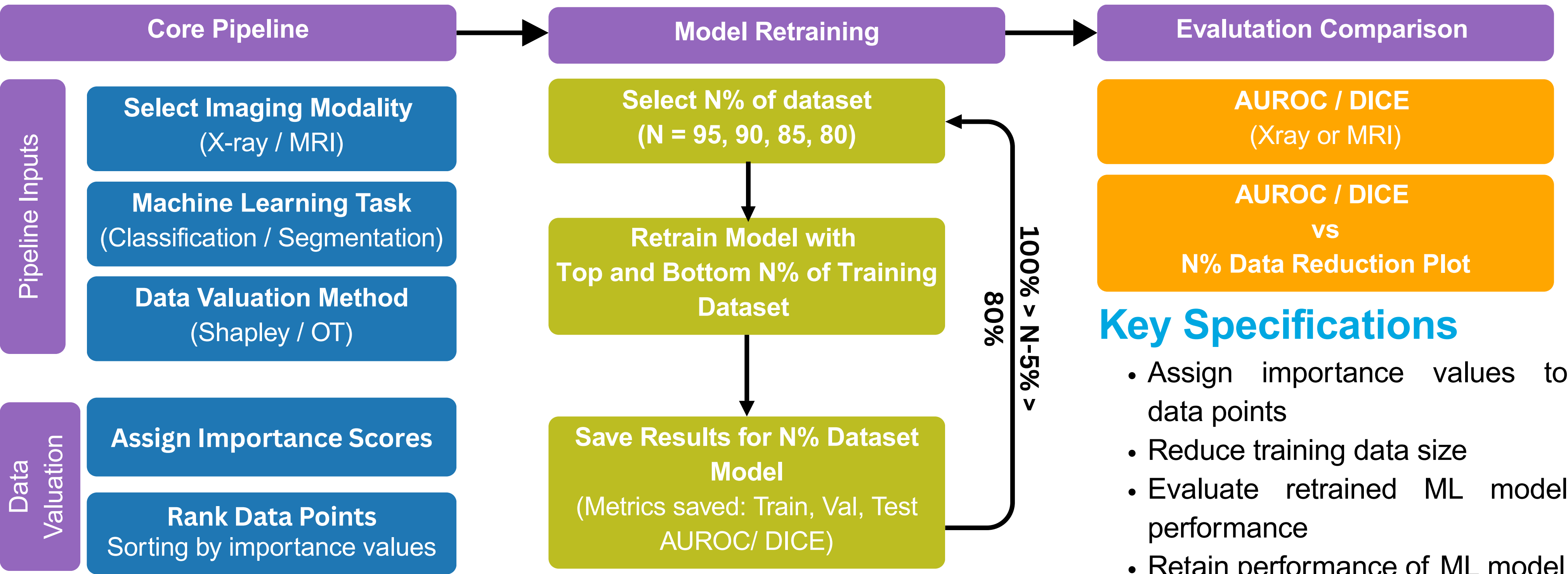
Shapley Values (k-NN approximation)

- **Match Local Neighborhood:** Measures how often a training sample's label matches its k nearest neighbor within feature space.
- **Importance Value:** Assign higher values to samples that consistently contribute to correct local labels.

Optimal Transport (OT)

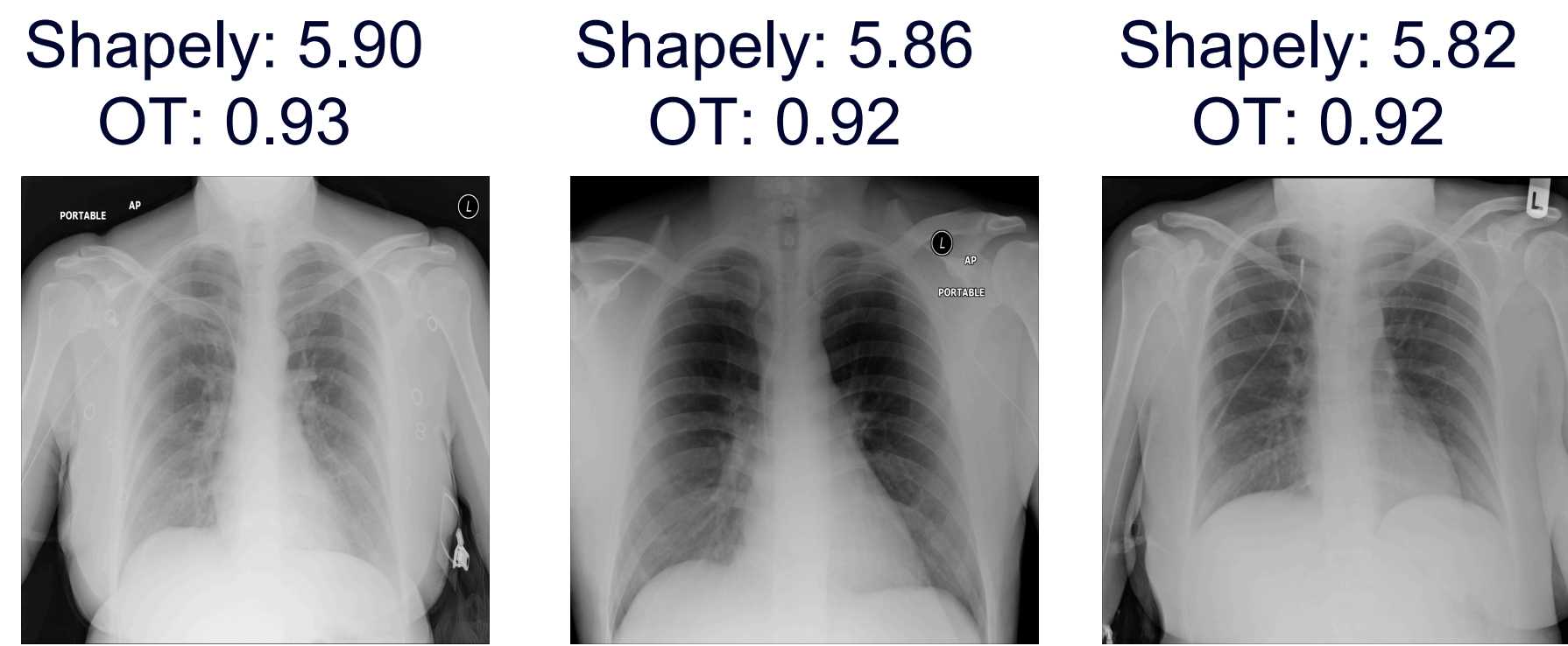
- **Map Two Distributions:** Calculates importance by computing a mathematical "transport map" that aligns the training set distribution with the validation set distribution.
- **Importance Value:** Assigns higher values to training samples that are structurally or statistically most similar to the target validation data distribution.

Pipeline

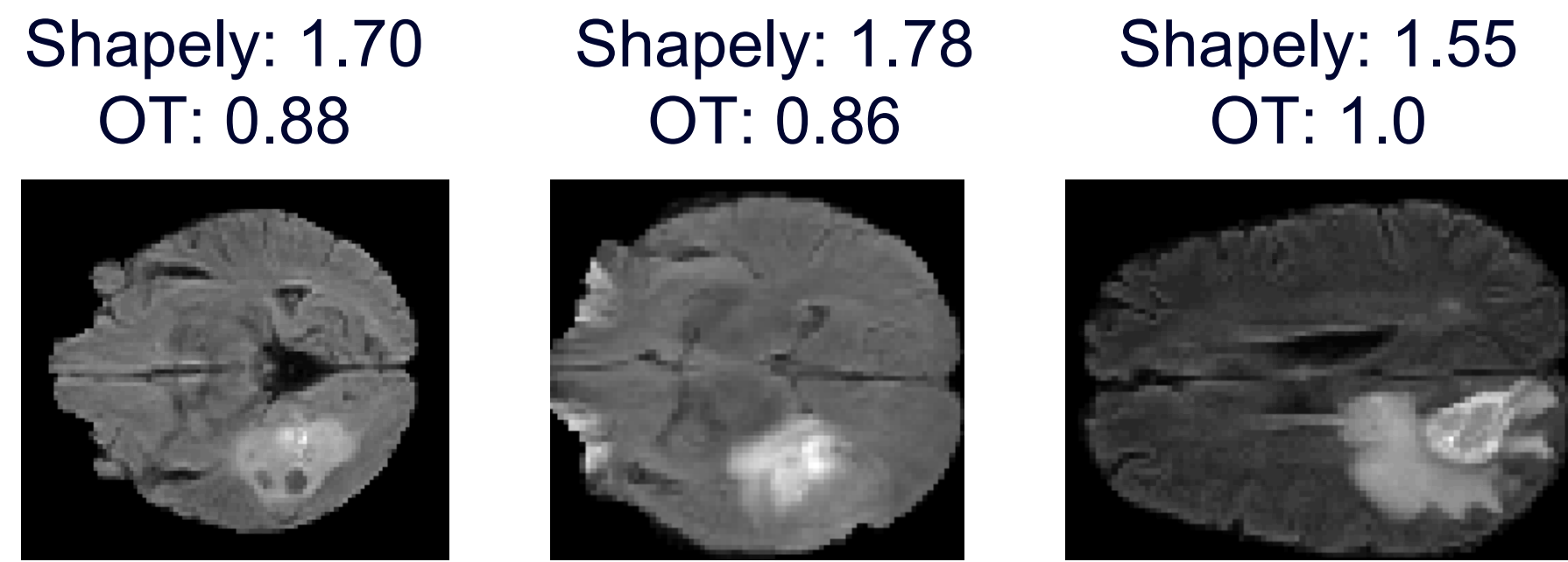


Results

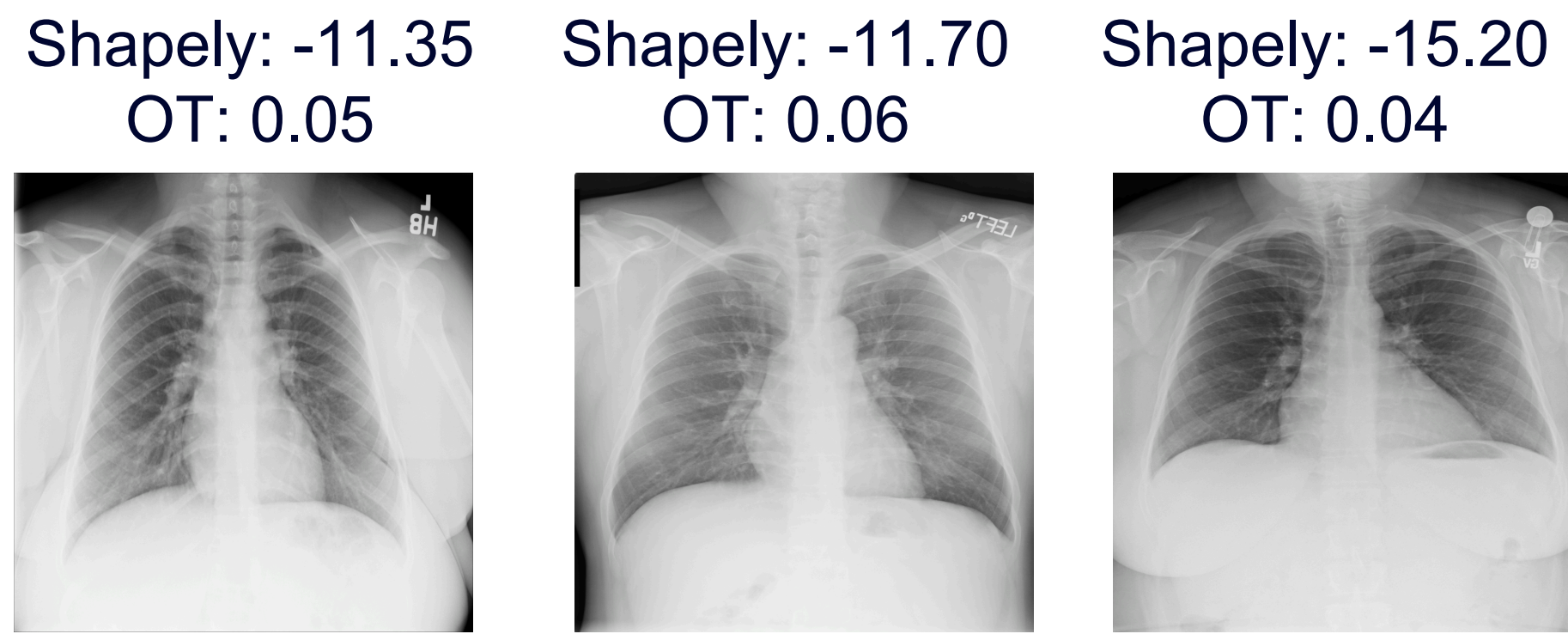
Top and Bottom Images for X-ray Classification



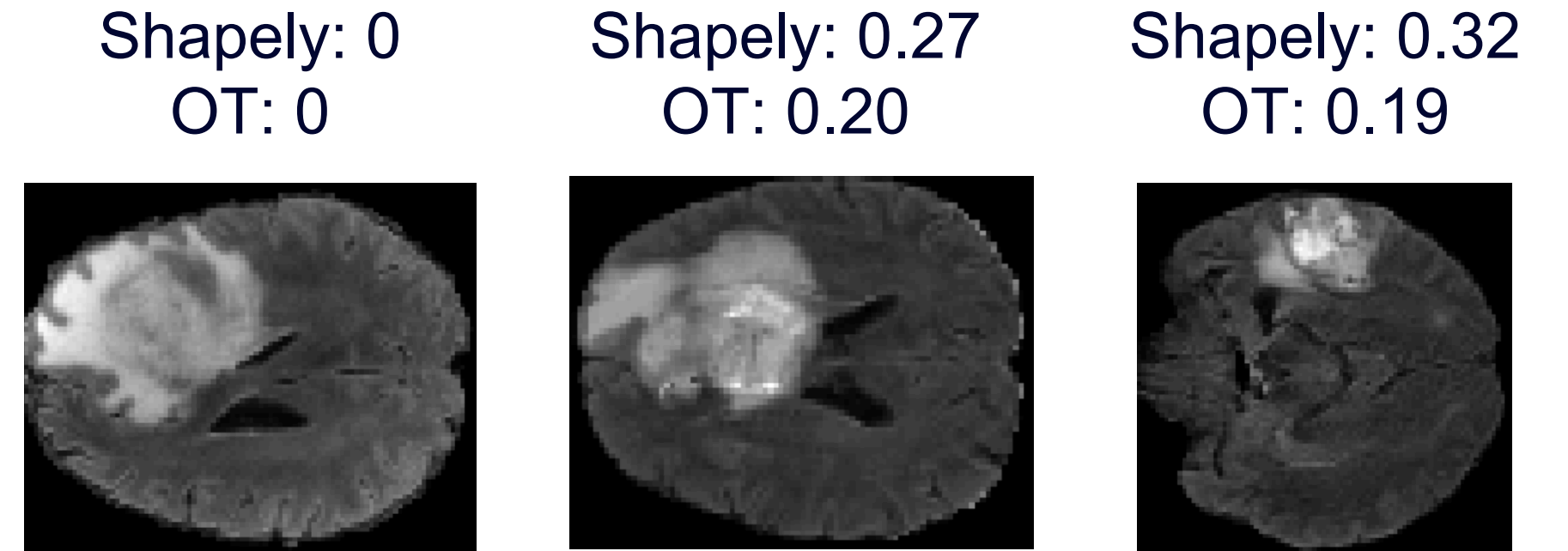
Top and Bottom Images for MRI Segmentation



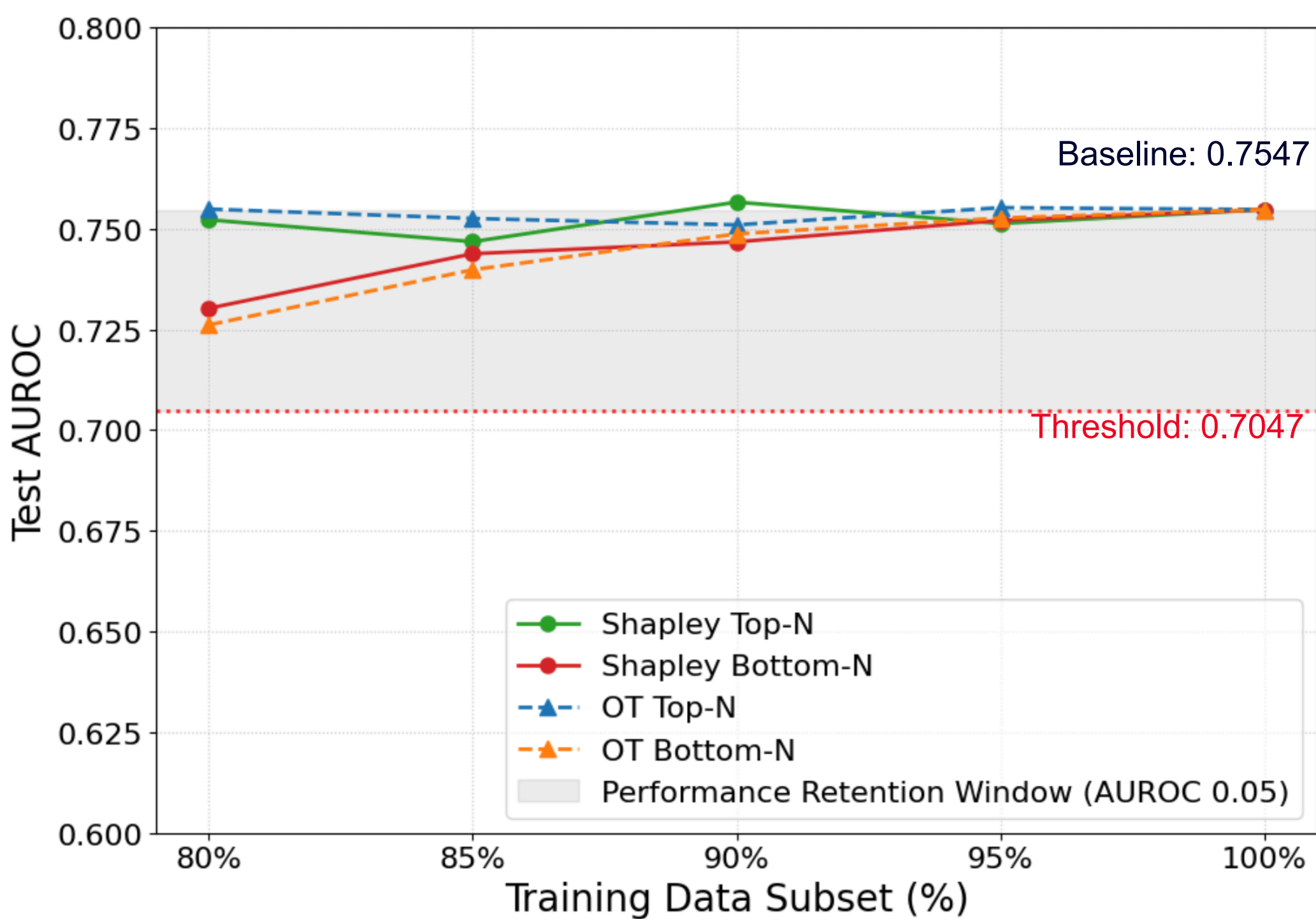
Size of Training Data: 896960



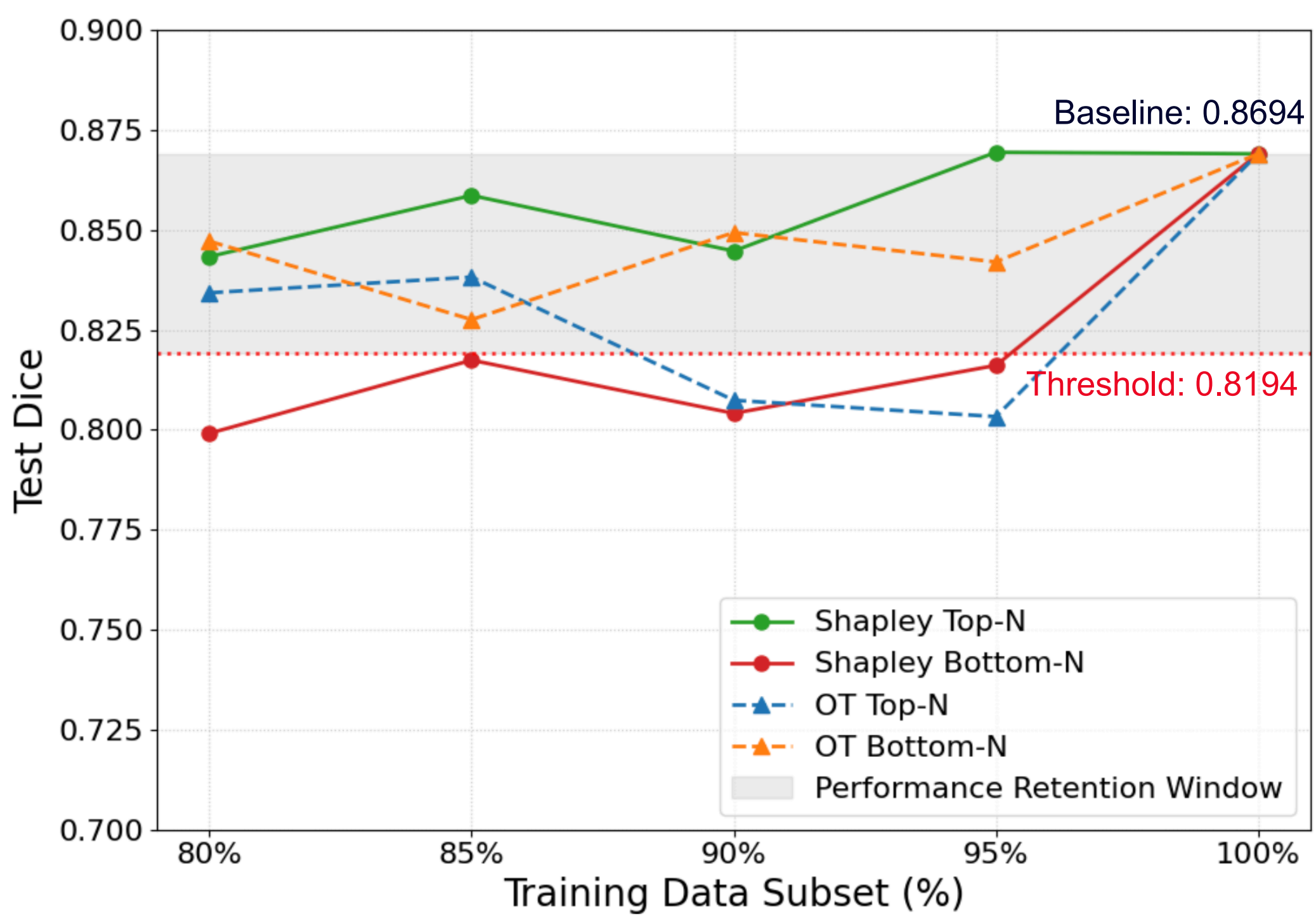
Size of Training Data: 900



X-ray Classification Test AUROC vs. Data Reduction

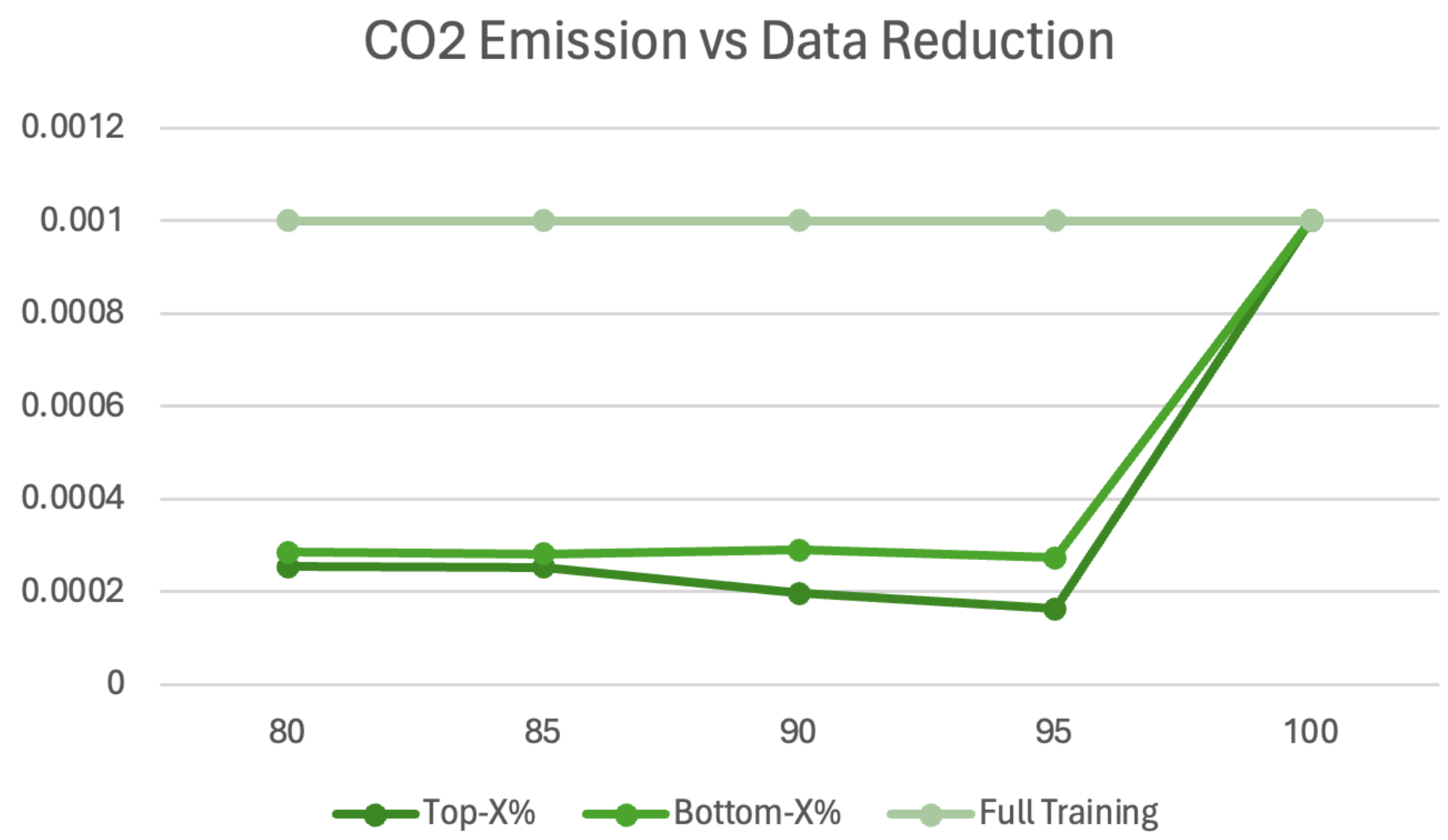


MRI Segmentation Test Dice vs. Data Reduction



Assumptions and Limitations

- Approximation method of Shapley was used to reduce the computational complexity of exact Shapely value
- Our findings are limited to the results output from the combinations of:
 - 2 DV methods
 - 2 ML architectures
 - 2 Modalities
- The data sets are from credible sources, but ground-truth annotations may still be imperfect
- Data set size and model run were not large enough to provide meaningful insights into CO2 emissions



Conclusion and Future Improvements

Our system:

- Effectively applies data valuation techniques (Shapley + OT) across X-ray Classification and MRI Segmentation
- Reduces training data size while retaining ML model performance
- Adheres to open-source and codebase tooling standards to encourage contributions from the ML community

Next steps:

- Further investigate Shapley and OT re-training experiments for both X-ray classification and MRI segmentation pipelines
- Explore additional DV methods (Topological Data Analysis) and medical imaging modalities
- Continue expanding environmental-impact analysis during data reduction using CodeCarbon to strengthen sustainability insights

Reference

- [1] S. Rajaraman, G. Zamzmi, F. Yang, Z. Liang, Z. Xue, and S. Antani, "Semantically redundant training data removal and deep model classification performance: A study with chest X-rays," Computerized Medical Imaging and Graphics, vol. 115, p. 102379, Jul. 2024, doi: <https://doi.org/10.1016/j.compmedimag.2024.102379>.
- [2] R. Girija, H. Singh, and G. Abirami, "Noise estimation in medical images based on fast discrete curvelet transform via wrapping," Results in Optics, vol. 19, p. 100820, Mar. 2025, doi: <https://doi.org/10.1016/j.rso.2025.100820>.
- [3] Staff, "Redundancy in medical imaging: Causes and fixes," Becker's Hospital Review | Healthcare News & Analysis, Oct. 13, 2014. <https://www.beckershospitalreview.com/quality/redundancy-in-medical-imaging-causes-and-fixes/> (accessed Apr. 02, 2026).
- [4] National Institute of Health, "NIH Chest X-rays," [www.kaggle.com. https://www.kaggle.com/datasets/nih-chest-xrays/data/data](https://www.kaggle.com/datasets/nih-chest-xrays/data/data)
- [5] Bionetworks, "BraTS 2023," Synapse.org, 2023. <https://www.synapse.org/Synapse:syn51156910/wiki/621282>

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